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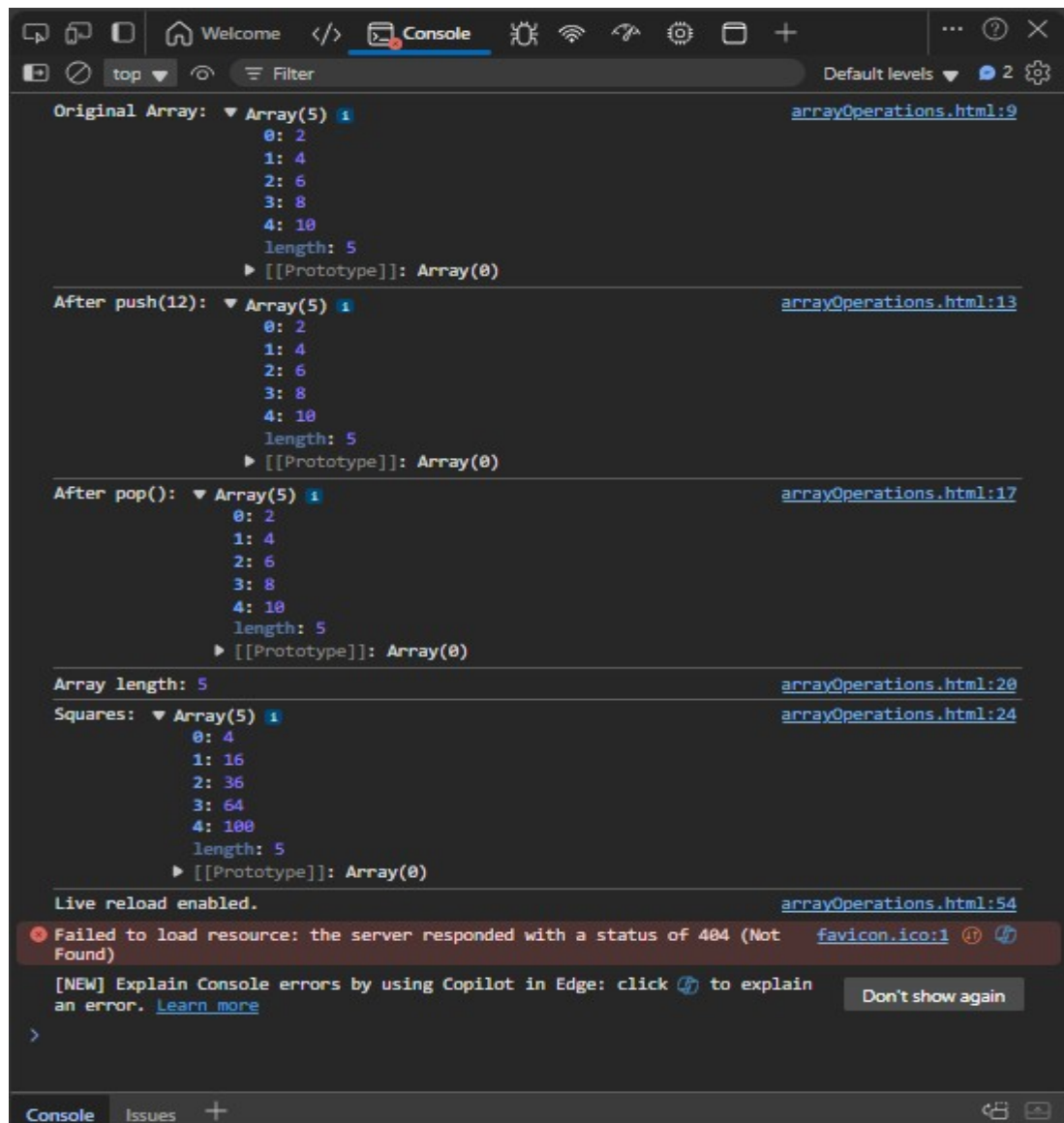
Sem : V

PRN : 23070521225

Code 1:

```
<> arrayOperations.html > ...
1  <!DOCTYPE html>
2  <head>
3    <title>Array Operations</title>
4  </head>
5  <body>
6    <script>
7      // Step 1: Create an array with 5 numbers
8      let numbers = [2, 4, 6, 8, 10];
9      console.log("Original Array:", numbers);
10
11     // Step 2: Add a new number using push()
12     numbers.push(12);
13     console.log("After push(12):", numbers);
14
15     // Step 3: Remove the last number using pop()
16     numbers.pop();
17     console.log("After pop():", numbers);
18
19     // Step 4: Print the length of the array
20     console.log("Array length:", numbers.length);
21
22     // Step 5: Use map() to get squares of numbers
23     let squares = numbers.map(num => num * num);
24     console.log("Squares:", squares);
25   </script>
26 </body>
27 </html>
```

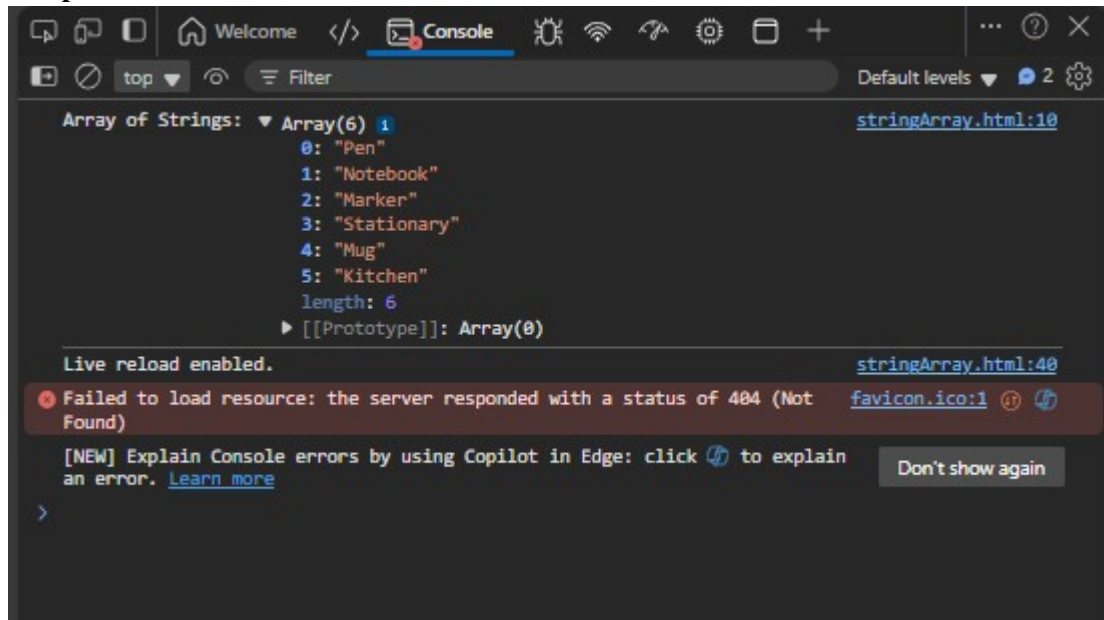
Output:



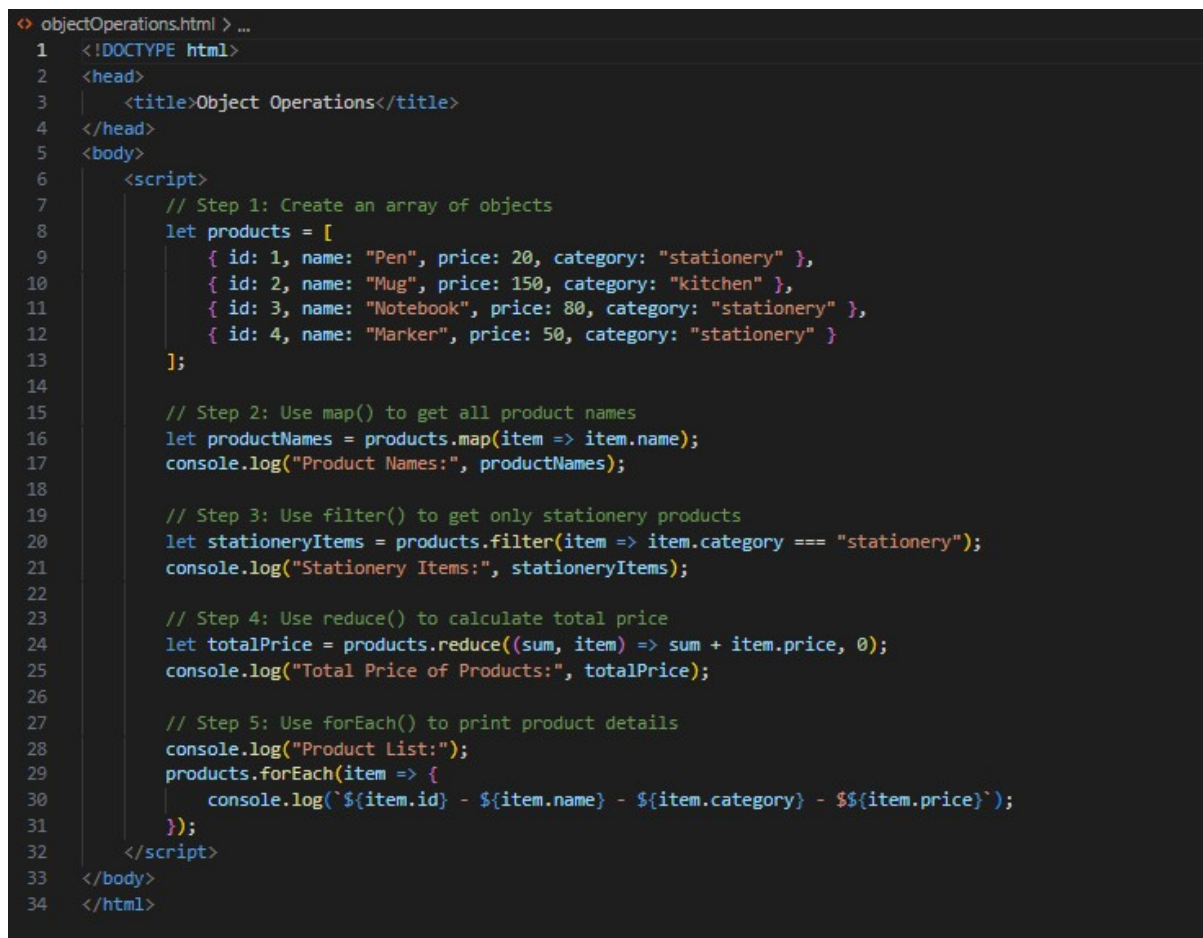
Code 2:

```
<> stringArray.html > ...
1  <!DOCTYPE html>
2  <head>
3  |   <title>Show String Array</title>
4  </head>
5  <body>
6  |   <script>
7  |     //Creating array of Strings
8  |     let arrStr = ["Pen", "Notebook", "Marker", "Stationary", "Mug", "Kitchen"];
9  |     //Displaying array of Strings
10 |     console.log("Array of Strings:", arrStr);
11 |   </script>
12 </body>
13 </html>
```

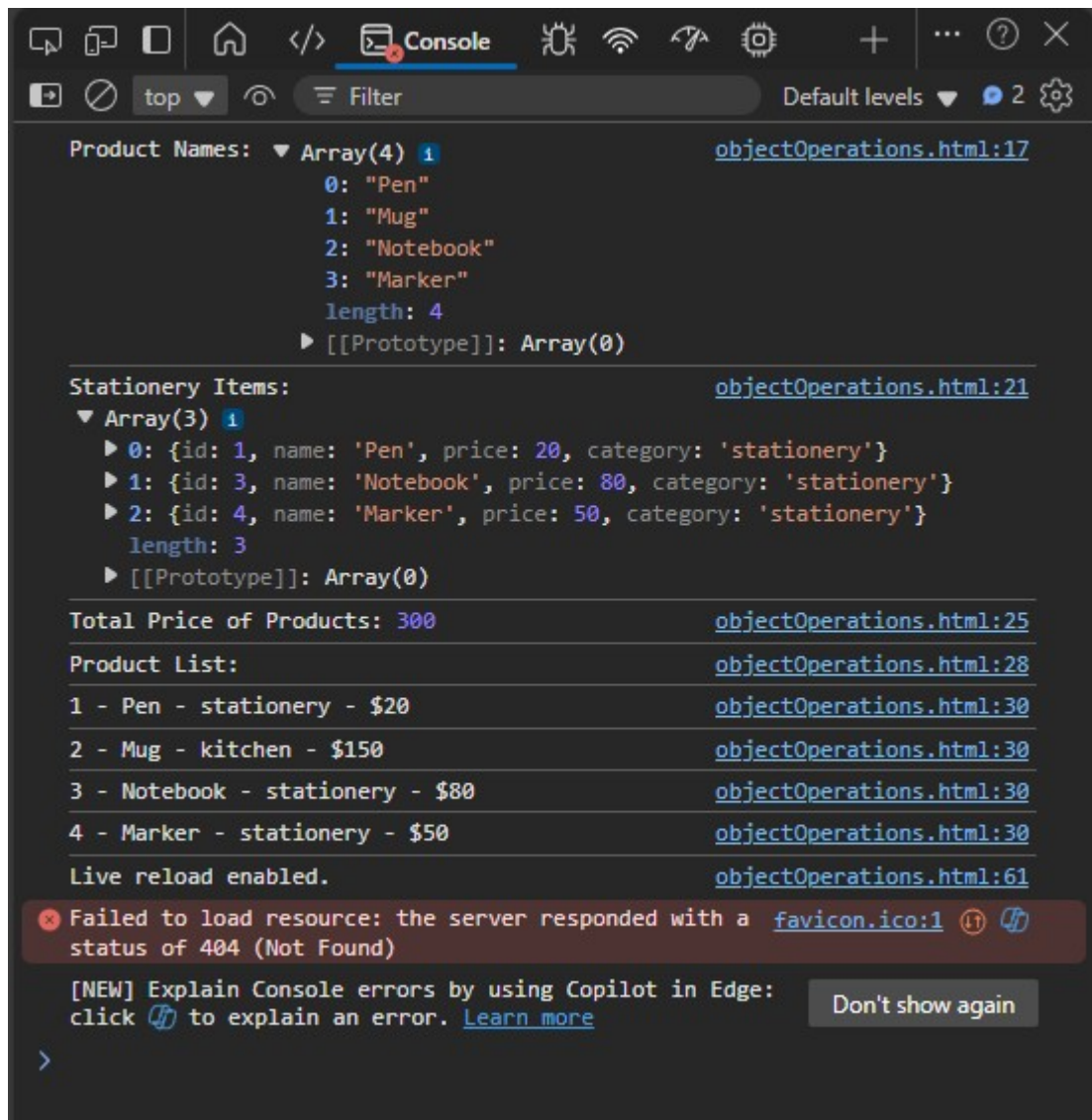
Output:



Code 3:



Output:



```
Product Names: ▼ Array(4) 1 objectOperations.html:17
  0: "Pen"
  1: "Mug"
  2: "Notebook"
  3: "Marker"
  length: 4
  ▶ [[Prototype]]: Array(0)

Stationery Items: objectOperations.html:21
▼ Array(3) 1
  ▶ 0: {id: 1, name: 'Pen', price: 20, category: 'stationery'}
  ▶ 1: {id: 3, name: 'Notebook', price: 80, category: 'stationery'}
  ▶ 2: {id: 4, name: 'Marker', price: 50, category: 'stationery'}
  length: 3
  ▶ [[Prototype]]: Array(0)

Total Price of Products: 300 objectOperations.html:25

Product List: objectOperations.html:28
1 - Pen - stationery - $20 objectOperations.html:30
2 - Mug - kitchen - $150 objectOperations.html:30
3 - Notebook - stationery - $80 objectOperations.html:30
4 - Marker - stationery - $50 objectOperations.html:30

Live reload enabled. objectOperations.html:61

❌ Failed to load resource: the server responded with a status of 404 (Not Found)
  favicon.ico:1
[NEW] Explain Console errors by using Copilot in Edge:
click to explain an error. Learn more
Don't show again
```

Code 4:

```
<> task1.html > ...
1  <!-- Create an array of student ages and use push() to add a new age.
2  Use pop() to remove the last age and display the array.
3  Print the length of the array.
4  Use map() to calculate double of each age.
5  Challenge: Use map() to find the cube of each element.      -->
6
7  <!DOCTYPE html>
8  <head>
9  |   <title>Array Operations</title>
10 </head>
11 <body>
12 |   <script>
13 |       //Create an array of student ages
14 |       let studentAges = [15, 16, 17, 18];
15 |       console.log("Initial Ages:", studentAges);
16 |
17 |       //Use push() method to add a new age
18 |       studentAges.push(19);
19 |       console.log("After push:", studentAges);
20 |
21 |       //Use pop() method to remove the last age
22 |       studentAges.pop();
23 |       console.log("After pop:", studentAges);
24 |
25 |       //Print the length of the array
26 |       console.log("Length of the array:", studentAges.length);
27 |
28 |       //Use map() method to calculate double of each age
29 |       let doubledAges = studentAges.map(age => age * 2);
30 |       console.log("Doubled Ages:", doubledAges);
31 |
32 |       //Challenge: Use map() method to find the cube of each element
33 |       let cubedAges = studentAges.map(age => age ** 3);
34 |       console.log("Cubed Ages:", cubedAges);
35 |   </script>
36 </body>
37 </html>
```

Output:



```
Initial Ages: ▼ Array(4) 1 task1.html:15
  0: 15
  1: 16
  2: 17
  3: 18
  length: 4
  ▶ [[Prototype]]: Array(0)

After push: ▼ Array(4) 1 task1.html:19
  0: 15
  1: 16
  2: 17
  3: 18
  length: 4
  ▶ [[Prototype]]: Array(0)

After pop: ▼ Array(4) 1 task1.html:23
  0: 15
  1: 16
  2: 17
  3: 18
  length: 4
  ▶ [[Prototype]]: Array(0)

Length of the array: 4 task1.html:26

Doubled Ages: ▼ Array(4) 1 task1.html:30
  0: 30
  1: 32
  2: 34
  3: 36
  length: 4
  ▶ [[Prototype]]: Array(0)

Cubed Ages: ▼ Array(4) 1 task1.html:34
  0: 3375
  1: 4096
  2: 4913
  3: 5832
  length: 4
  ▶ [[Prototype]]: Array(0)

Live reload enabled. task1.html:64

Failed to load resource: the server responded with a status of 404 (Not Found) favicon.ico:1
[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. Learn more
Don't show again
```

Code 5:

```

task2.html > ...
1  <!-- Create an array of objects for students with properties: { id, name, marks }.
2  Use map() to extract all student names.
3  Use filter() to select students with marks > 50.
4  Use reduce() to calculate the average marks.
5  Use forEach() to print a report card format.      -->
6
7  <!DOCTYPE html>
8  <head>
9    <title>Object Operations</title>
10 </head>
11 <body>
12   <script>
13     //Create an array of student objects
14     let students = [
15       {id : 1, name : "Aryan", marks : 94},
16       {id : 2, name : "Riya", marks : 87},
17       {id : 3, name : "Karan", marks : 67},
18       {id : 4, name : "Meena", marks : 42},
19       {id : 5, name : "Amit", marks : 35}
20     ];
21
22     //Use map() method to extract all student names
23     let studentNames = students.map(student => student.name);
24     console.log("Student Names:", studentNames);
25
26     //Use filter() method to select students with marks>50
27     let passedStudents = students.filter(student => student.marks>50);
28     console.log("Students with marks > 50:", passedStudents);
29
30     //Use reduce() method to calculate the average marks
31     let totalMarks = students.reduce((sum, student) => sum + student.marks, 0);
32     let averageMarks = totalMarks / students.length;
33     console.log("Average Marks:", averageMarks);
34
35     //Use forEach() method to print a report card format
36     console.log("\n Report Card:");
37     students.forEach(student => {
38       console.log(`ID: ${student.id} | Name: ${student.name} | Marks: ${student.marks}`);
39     });
40   </script>
41 </body>
42 </html>

```

Output:

```
Student Names: ▼ Array(5) 1 task2.html:24
  0: "Aryan"
  1: "Riya"
  2: "Karan"
  3: "Meena"
  4: "Amit"
  length: 5
  ▶ [[Prototype]]: Array(0)

Students with marks > 50: task2.html:28
▼ Array(3) 1
  ▶ 0: {id: 1, name: 'Aryan', marks: 94}
  ▶ 1: {id: 2, name: 'Riya', marks: 87}
  ▶ 2: {id: 3, name: 'Karan', marks: 67}
  length: 3
  ▶ [[Prototype]]: Array(0)

Average Marks: 65 task2.html:33

Report Card: task2.html:36
ID: 1 | Name: Aryan | Marks: 94 task2.html:38
ID: 2 | Name: Riya | Marks: 87 task2.html:38
ID: 3 | Name: Karan | Marks: 67 task2.html:38
ID: 4 | Name: Meena | Marks: 42 task2.html:38
ID: 5 | Name: Amit | Marks: 35 task2.html:38
Live reload enabled. task2.html:69
>
```